

Library walking access

Phoenix, Arizona, USA

5%

of residents can reach a library within a 15-minute walk (slope-aware travel time)

1,446,730

people are beyond that 15-minute standard

98%

of residents with an ambulatory difficulty are beyond a 15-minute walk

145 min

is the median walk faced by the worst-served of that group — before any slope penalty at all

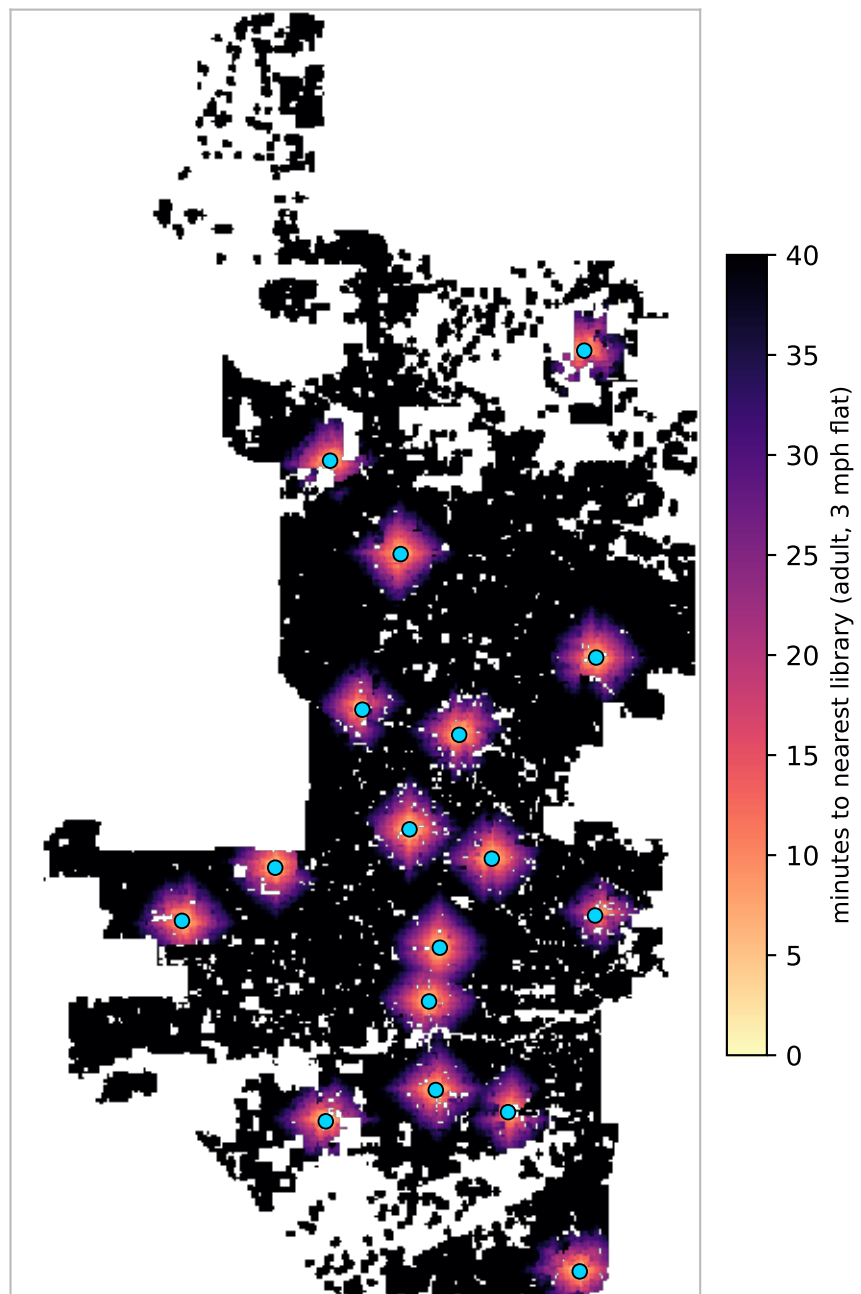
Scope 17 library locations · 1,002 km² analysed · 1,522,291 residents

Terrain elevation 289–820 m · 1.9% of walk segments steeper than the 5% accessible-route grade
Standard 15-minute neighbourhood goal; 3 mph on the flat = 0.75 mi

Sources OpenStreetMap walk network · US Census ACS 5-year 2023 · USGS 3DEP elevation

Where the walk is long

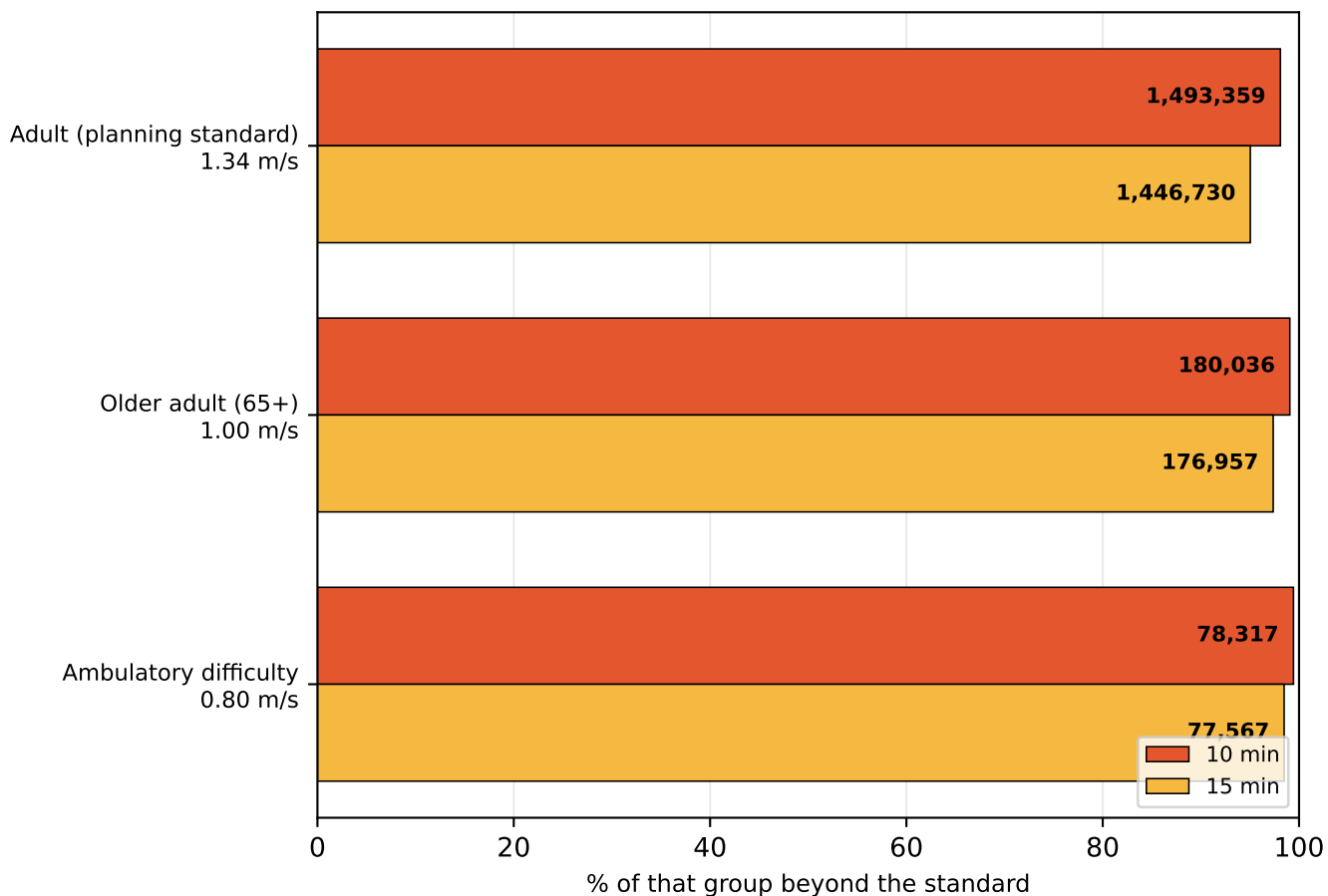
Travel time to the nearest library, on foot, accounting for slope



Median walk is 53 minutes. Blue dots are the 17 branches. Darker areas are further in time, not distance — a steep half-mile costs more than a flat one. Areas outside the analysed land mask (open water, and cells more than 250 m from any mapped pedestrian way) are blank.

A time standard is not one distance

The same policy sentence describes a different city for each kind of walker

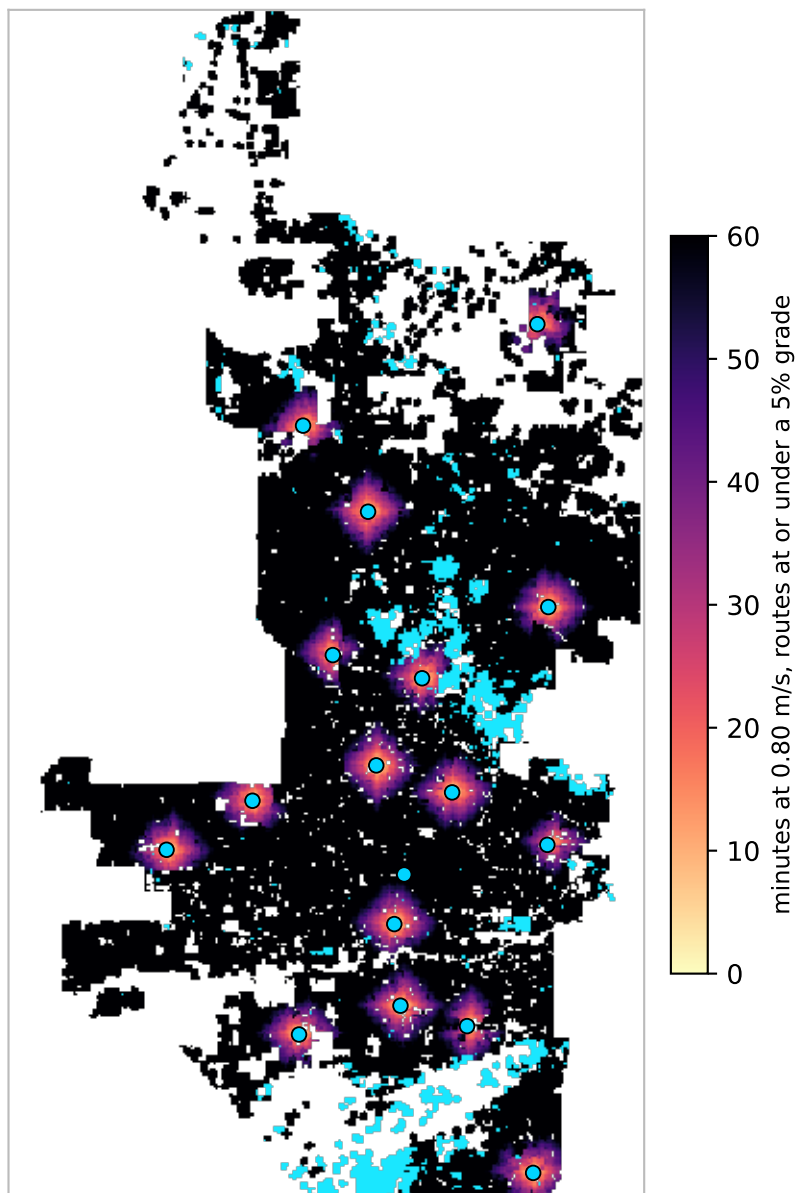


profile	group	10 min	15 min	no route
Adult (planning standard)	1,522,291	1,493,359 (98%)	1,446,730 (95%)	0
Older adult (65+)	181,742	180,036 (99%)	176,957 (97%)	0
Ambulatory difficulty	78,765	78,317 (99%)	77,567 (98%)	1,063

Each row is measured against its own population, not a share of the city. Walking speeds are planning defaults: 3 mph adult, 1.00 m/s for 65+, 0.80 m/s with an ambulatory difficulty.

Slope, and who it excludes

Measured against the 1:20 (5%) grade an accessible route is designed to



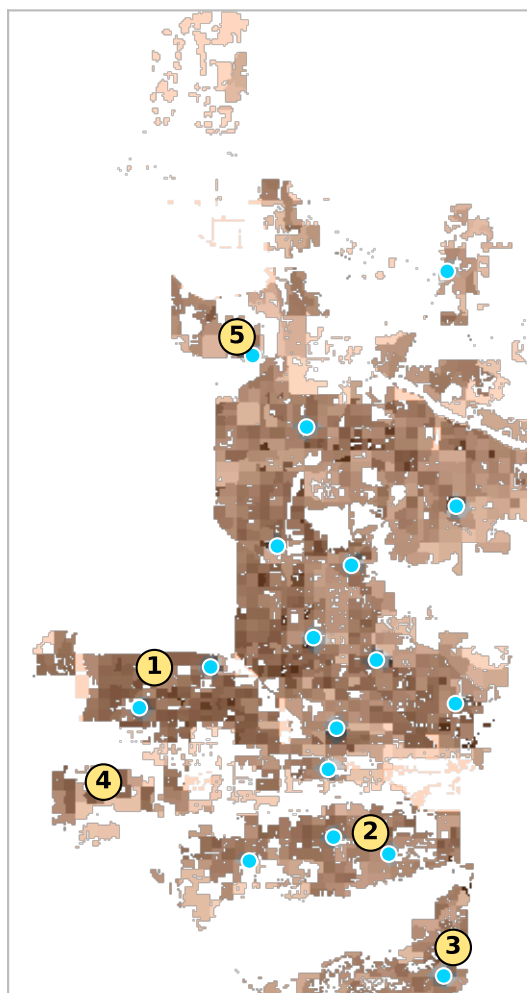
Cyan marks land with no route to any library that stays within a 5% grade: 1,063 residents with an ambulatory difficulty, 1% of that group. 1.9% of walk segments are steeper than that.

That count is sensitive to the threshold — it ranges 79–1,628 across cutoffs from 4% to 15% — so it should not be read as a precise figure. Two things underneath it are not sensitive.

These are remote places before grade is considered at all: with no slope penalty whatsoever the same cohort still faces a median 145-minute walk, and 87% over an hour. And the burden falls almost entirely on this profile — grade moves the adult figure about two points, against 0 points here.

Underserved pockets and best sites

Contiguous residential areas beyond a 15-minute walk, ranked by residents affected



#	residents	car-free HH	area	worst walk	site gains
1	1,083,825	30,195	481.3 km ²	198 min	12,582
2	146,326	2,616	72.0 km ²	148 min	7,226
3	61,004	730	29.4 km ²	122 min	5,463
4	41,176	44	18.6 km ²	213 min	6,905
5	27,700	18	17.1 km ²	128 min	4,368
6	19,728	245	15.1 km ²	155 min	4,074

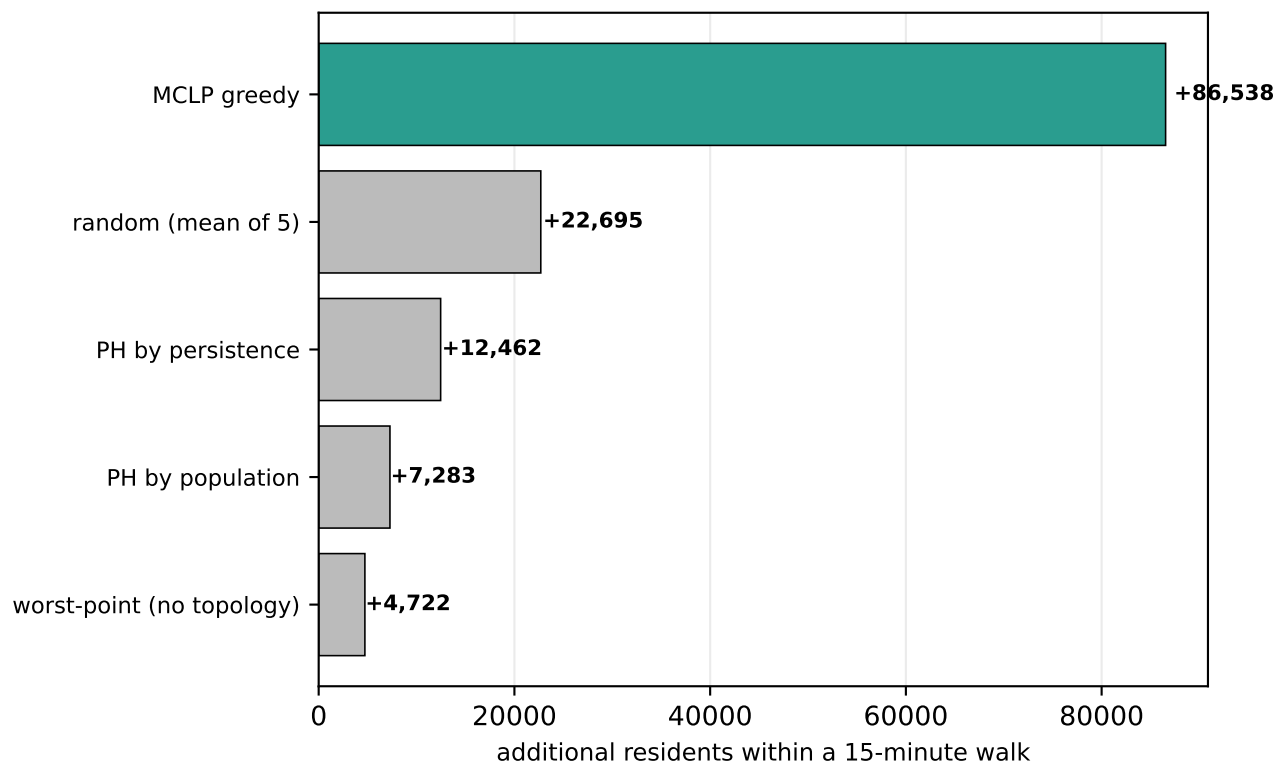
Numbered markers are the best available branch site inside each pocket — the location that brings the most residents within 15 minutes — chosen from habitable land only, so parks, port terminals and airfields are never proposed.

These are not real parcels.

A marker means “somewhere around here would help most”, resolved to a 150 m grid. It carries no assessment of zoning, ownership, lot size, acquisition cost, or whether anything is buildable on the site. Treat each one as a search area for a siting study, not a candidate address.

Where should the next 8 branches go?

Siting strategies scored on residents brought within 15 minutes



Greedy maximal-covering location (MCLP) adds 86,538 residents with 8 branches, raising coverage from 5% to 11%. It is scored here against topological and trivial alternatives; see the repository for the full comparison. Candidate sites are inhabited cells on a 450 m grid.

Method and caveats

Walk network from OpenStreetMap, retaining all connected components. Travel time uses Tobler's hiking function renormalised to each profile's flat speed; the mobility profile additionally treats segments steeper than 5% as impassable. Population is ACS 2023 block-group data rasterised to a 150 m grid; poverty, vehicle access and ambulatory difficulty are tract-level and therefore coarser.

Known limits: sidewalk quality, curb ramps, crossing delay and transit are not modelled, so the mobility figures remain optimistic even with slope. Walking speeds are literature defaults rather than locally observed. Travel time is one-way; a downhill outbound trip is uphill on return. Facility sets are branch locations only and take no account of opening hours or capacity.